3.7 曲率

弧微分

在数学的发展过程中，数学家想求一个弧（或曲线）的长度，在这个过程中发明了曲线积分，当然这是我们后面要学习的内容，这里面我们不过多涉及。

我们先简单说明一些与弧相关基本的概念。

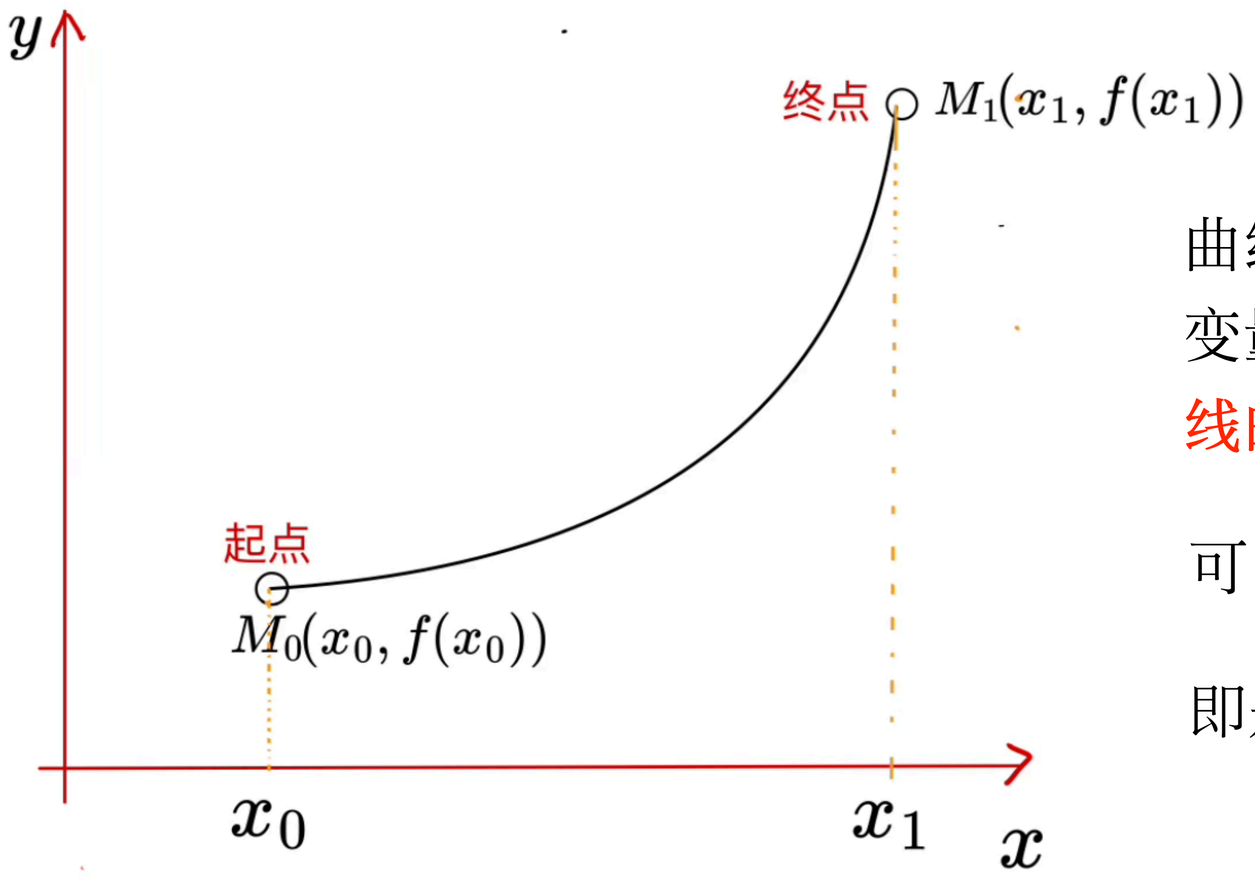
弧微分

曲线的方向或弧的方向

$y=f(x)$连续且曲线没有重点

如图，设曲线 $y=f(x)$ 是一条简单的连续的曲线；称 $M_0(x_0,f(x_0))$ 为曲线的起点；

称 $M_1(x_1,f(x_1))$ 为曲线的终点；

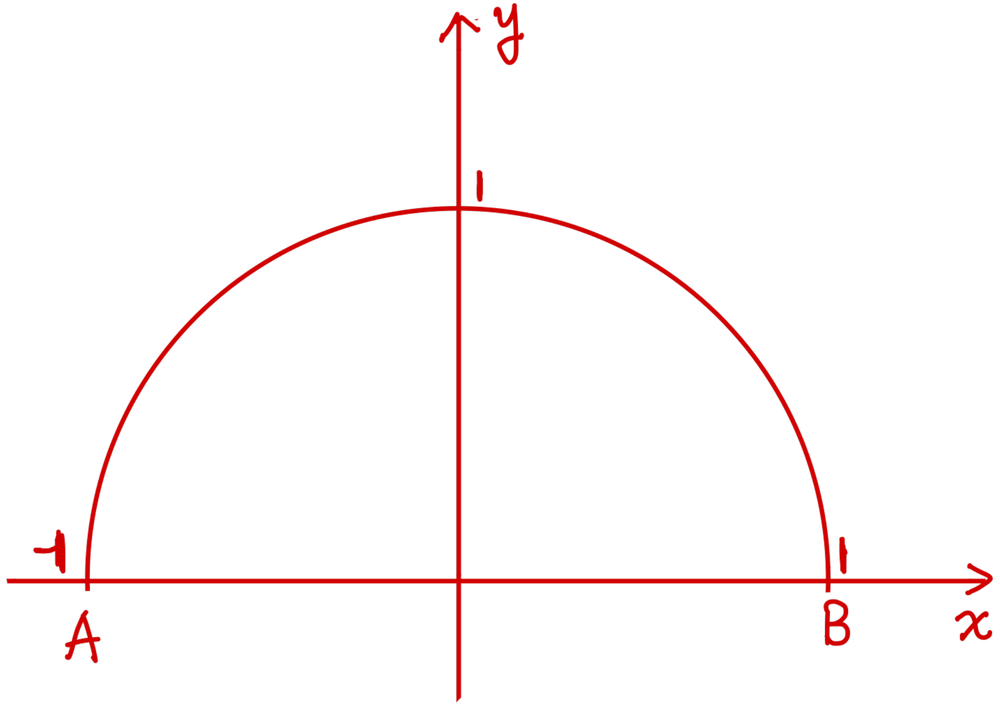


曲线上的点依变量 x 增加的顺序排列，变量 x 大的点在后面，这样就确定了曲线的正方向或弧的正方向

可以证明曲线的方向与曲线的方程形式无关

即是说，曲线的方向是一个几何上的概念.

弧微分



比如，上半平面中的单位圆

$$y=\sqrt{1-x^2},-1\le x\le 1$$

它还有参数方程：$\begin{cases} x=\cos\theta, \\ y=\sin\theta, \end{cases}$ 其中 $-\pi\le\theta\le\pi$.

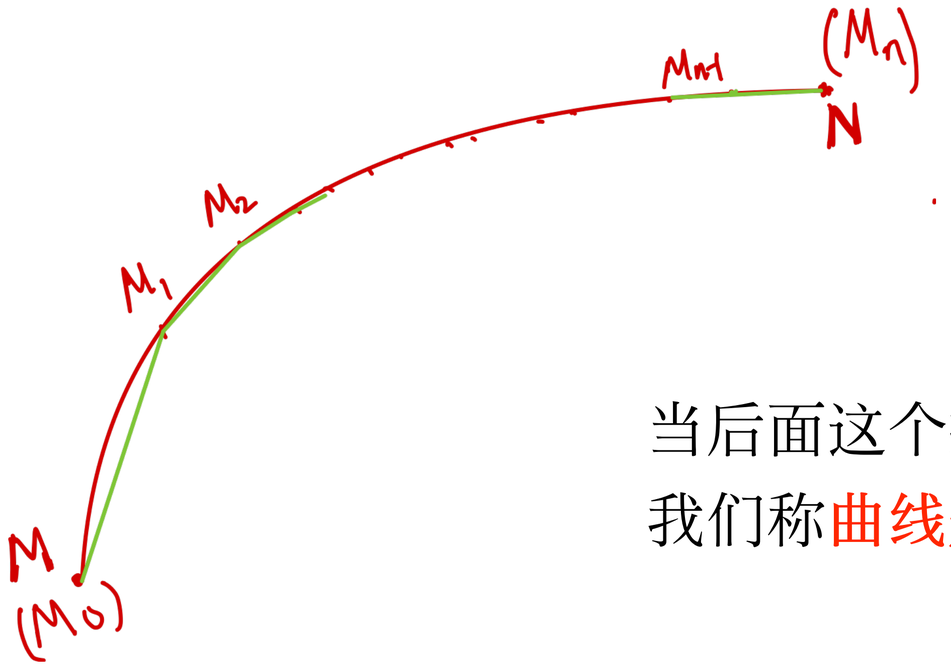
两种形式曲线的正方向都对应从 A 指向 B

或曲线由 A 到 B 的方向是正方向或正方向

弧微分

可求长度的曲线

并不所有的曲线都是可求长度的。那么哪些曲线可求长度的呢?



比如：左边的曲线，我们描出一些点 $M_0, M_1, M_2, \ldots, M_n$

显然：$\overset{\frown}{M_iM_{i+1}} > |M_iM_{i+1}|$

所以：$\overset{\frown}{MN} > \sum_{i=0}^{n} |M_iM_{i+1}|$

当后面这个拆线长的总和对于任意的 n 都有限或者有上界时，我们称**曲线是可求长度的**.

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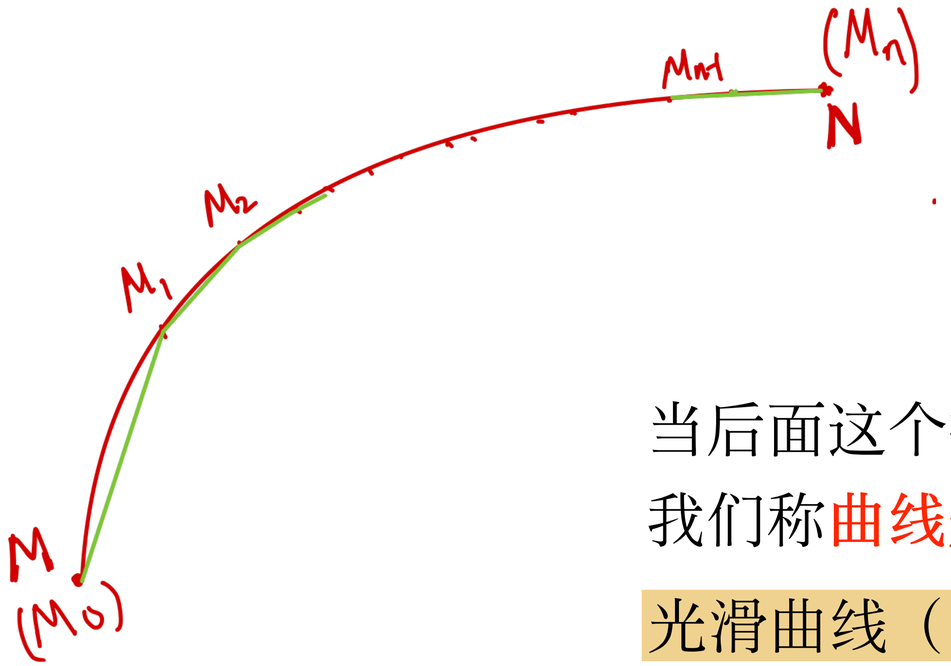

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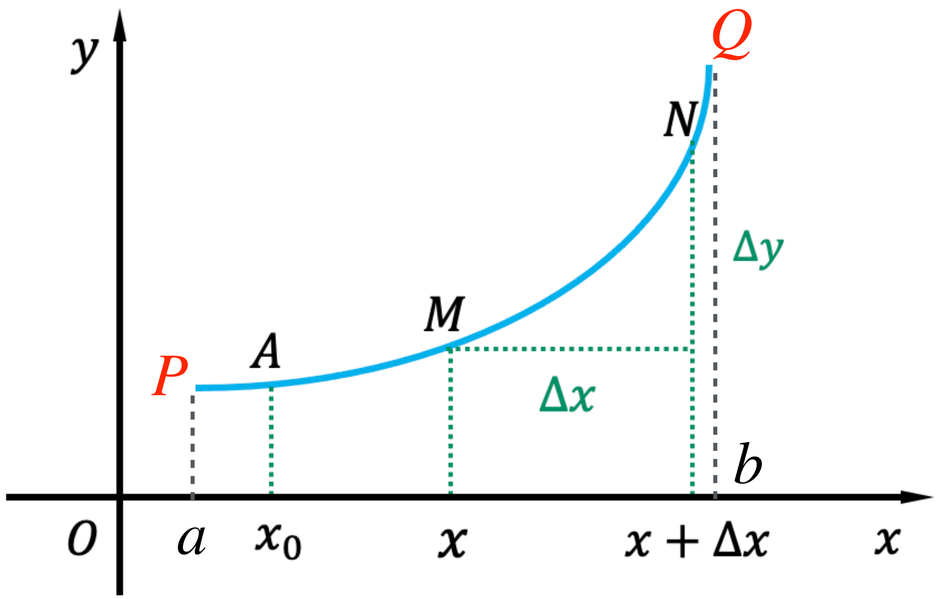
光滑曲线（导函数连续）是可求长度的曲线

分段光滑曲线是可求长度的曲线

弧微分

以后我们所考虑的曲线都是（分段）光滑的曲线。

如图，函数 $y=f(x)$ 在区间 (a,b) 内具有连续的导函数（曲线光滑）



曲线的正方向：从 P 点指向 Q 点

固定曲线上一点 $A(x_0,f(x_0))$,

设 M 为曲线上任意一点 $M(x,f(x))$,

考虑有向弧 $\overset{\frown}{AM}$ 的长度记为 s,

如果曲线 AM 的正方向与曲线 PQ的正方向一致，

则规定 s 取正值，否则取负值

显然，s 与 x 有关，且不同的 x 对应不同的 s

即 s 是 x 的函数记为 $s(x)$

显然 $s(x)$ 是关于 x 的增函数.

弧微分

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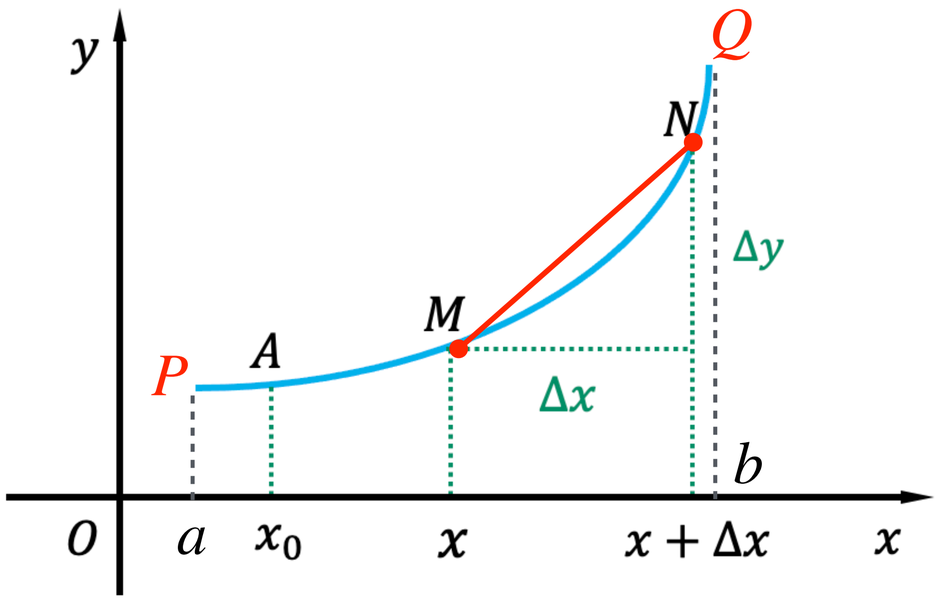

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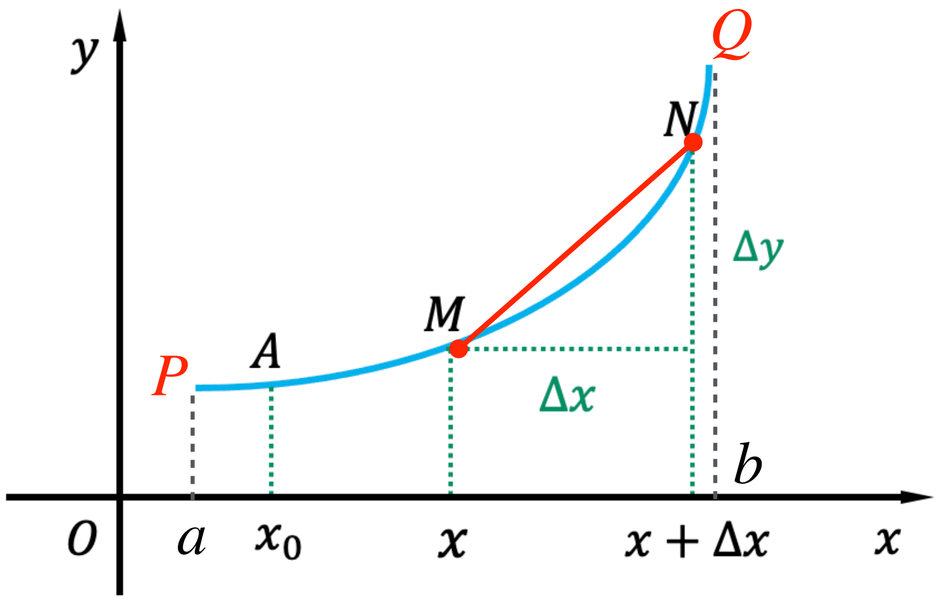

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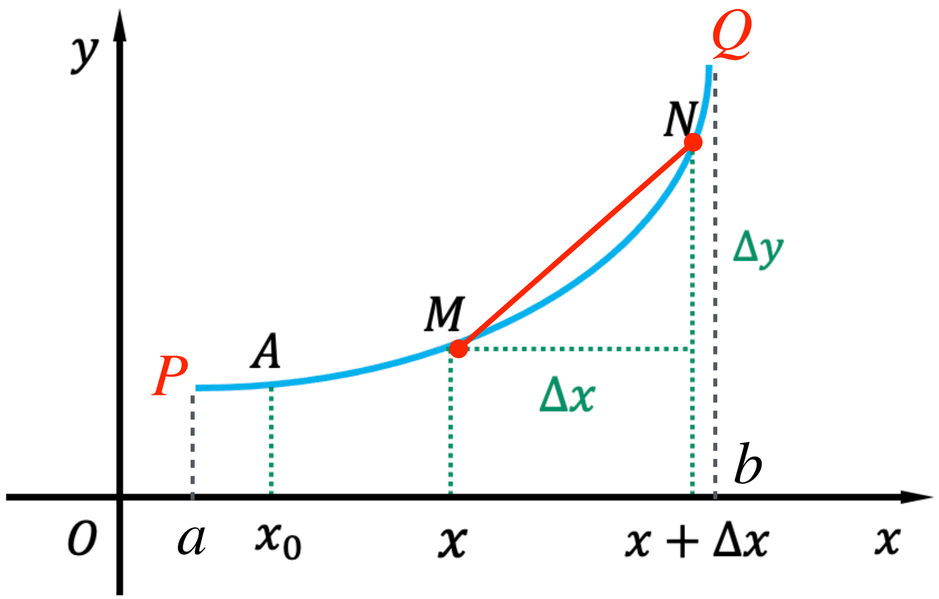

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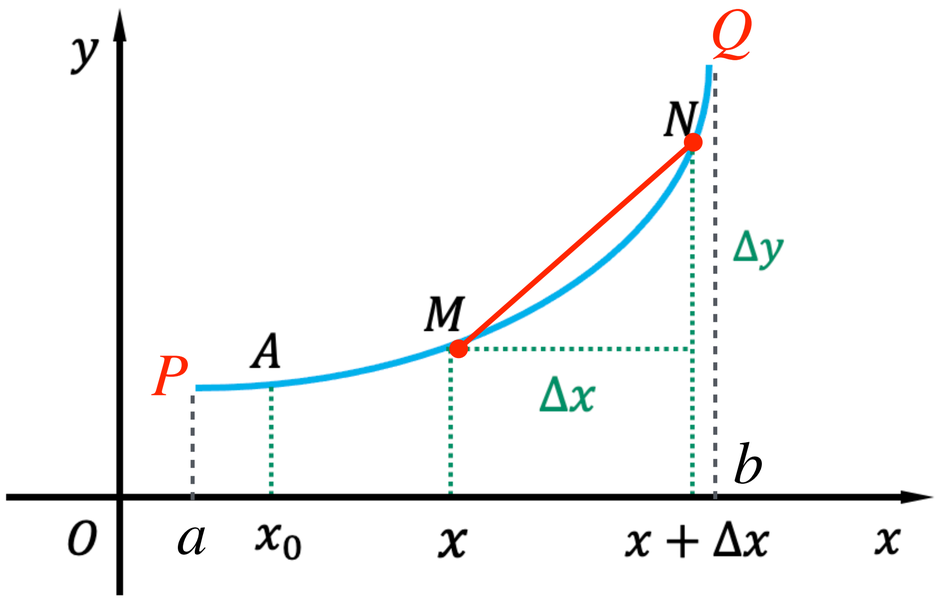

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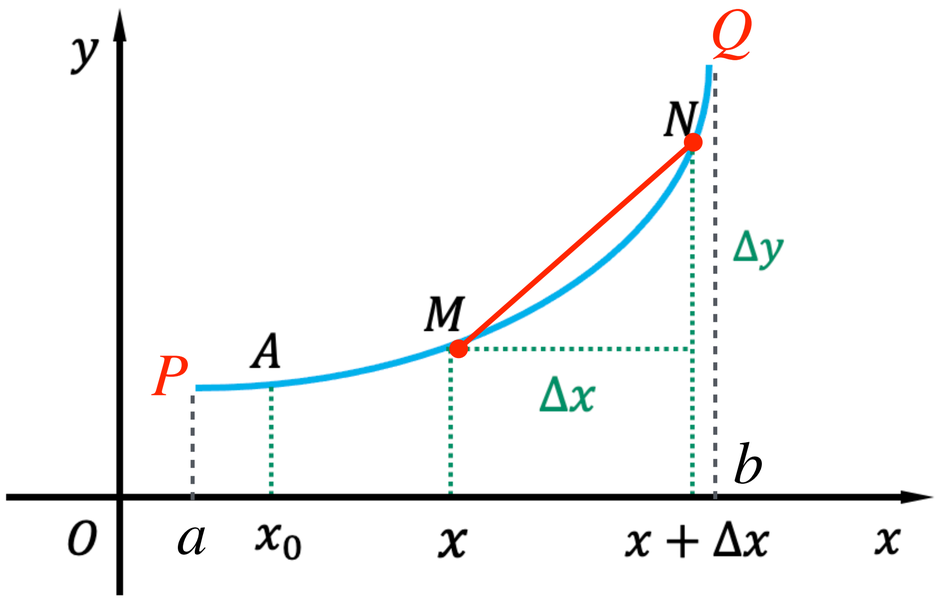

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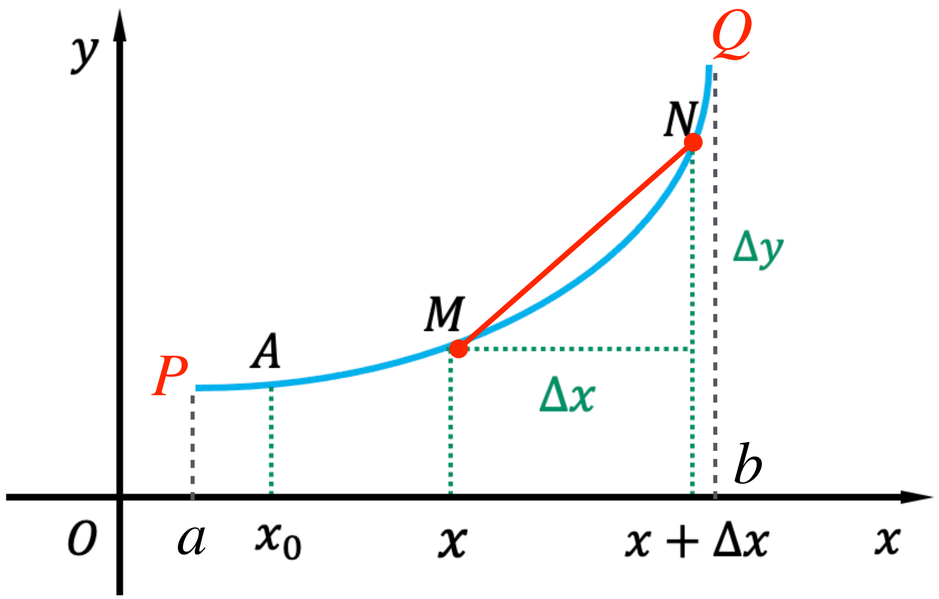

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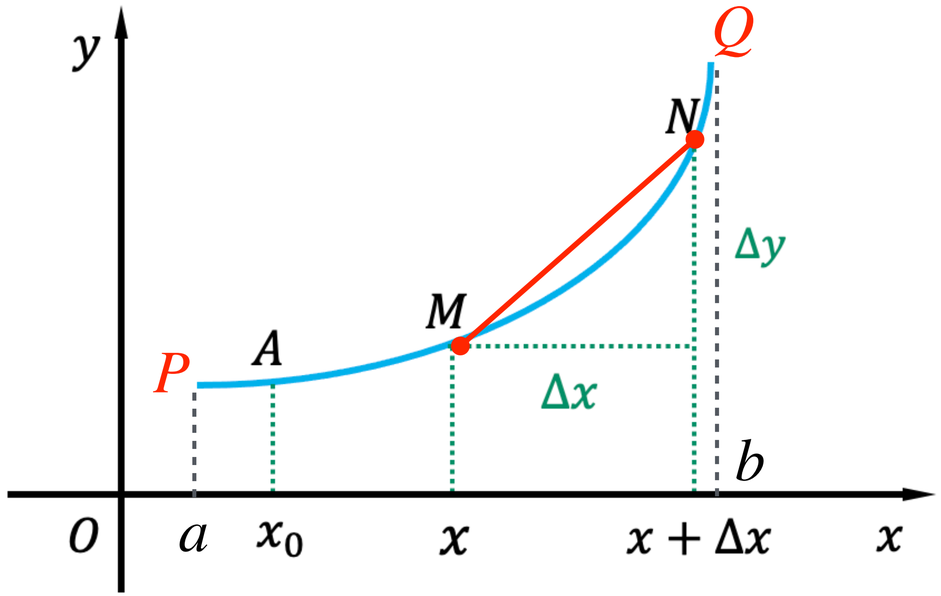

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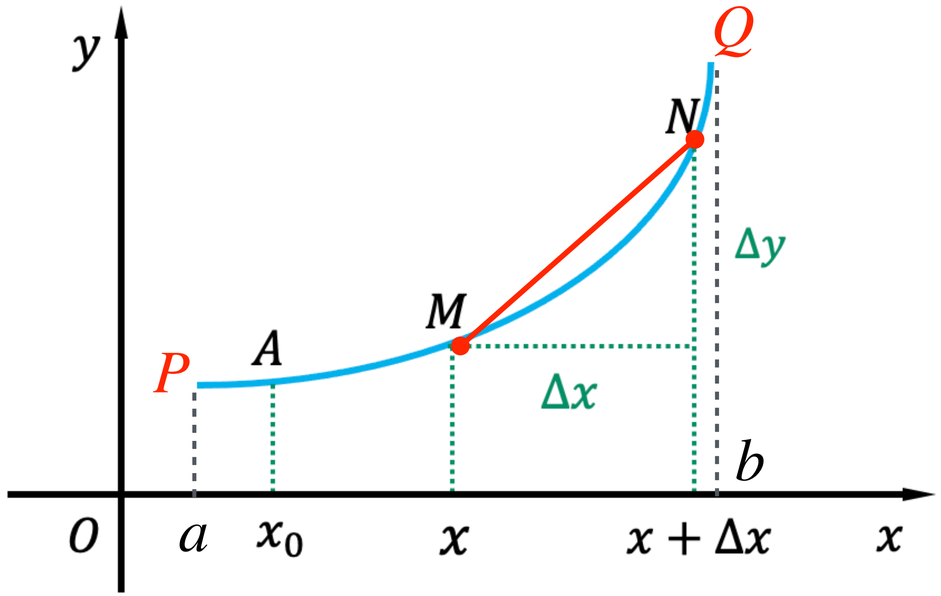

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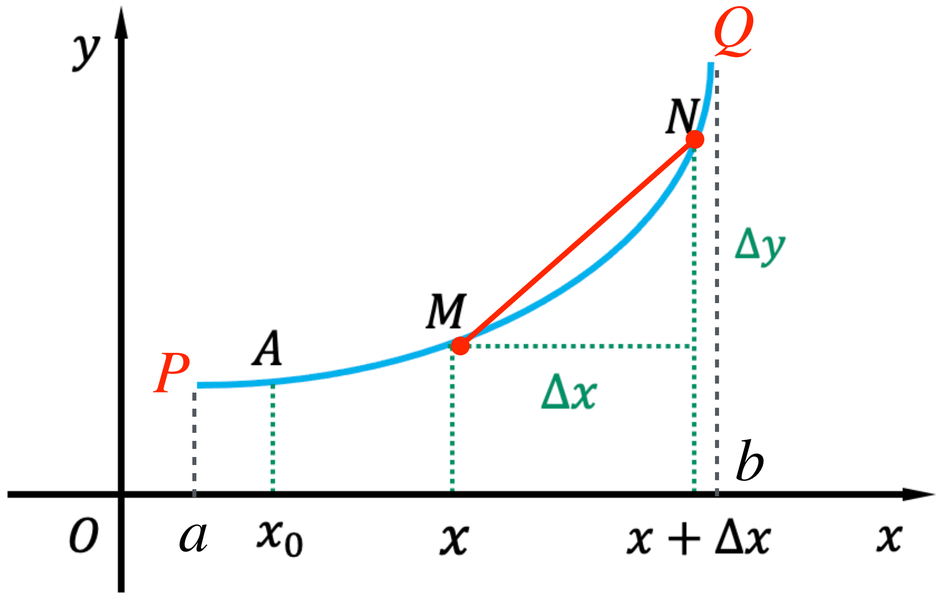

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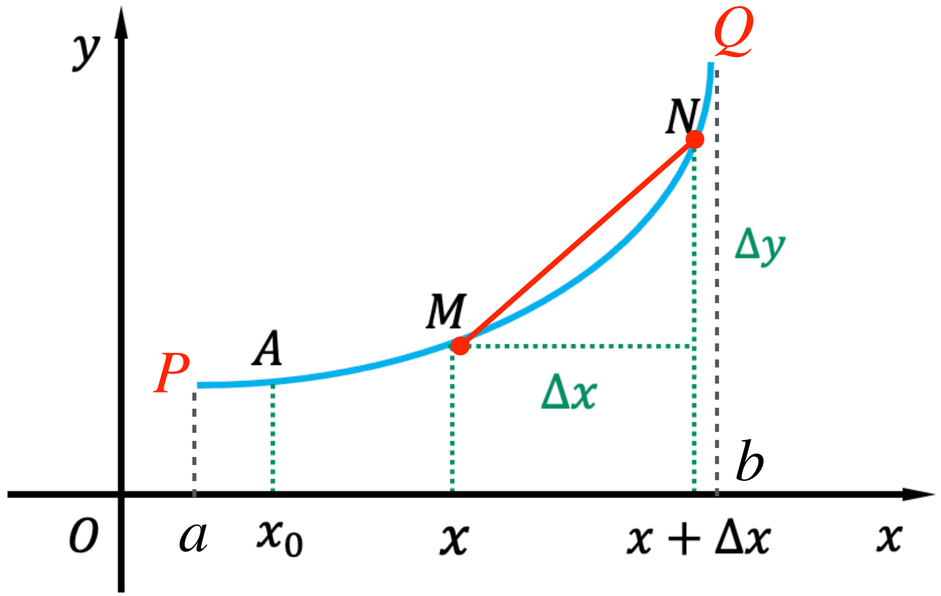

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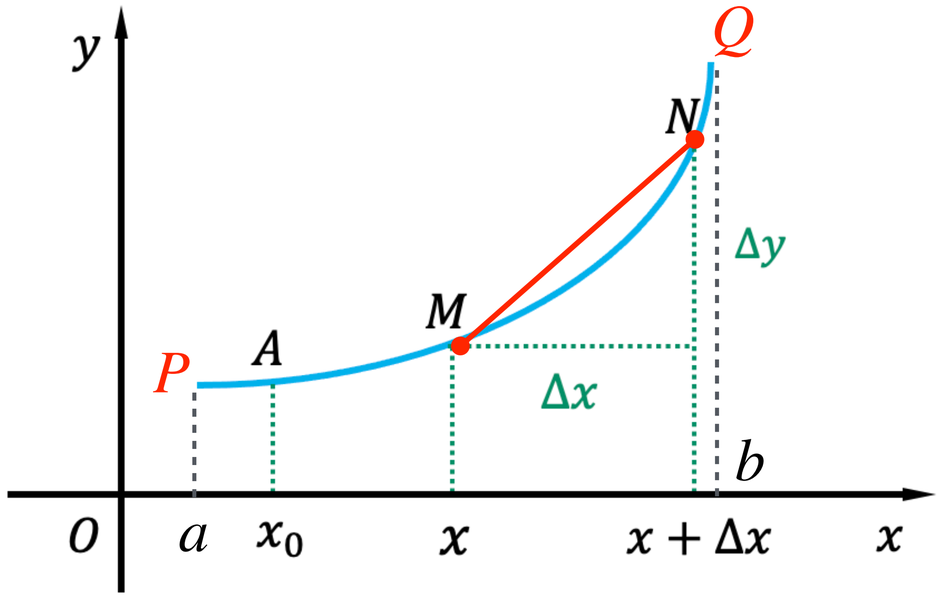

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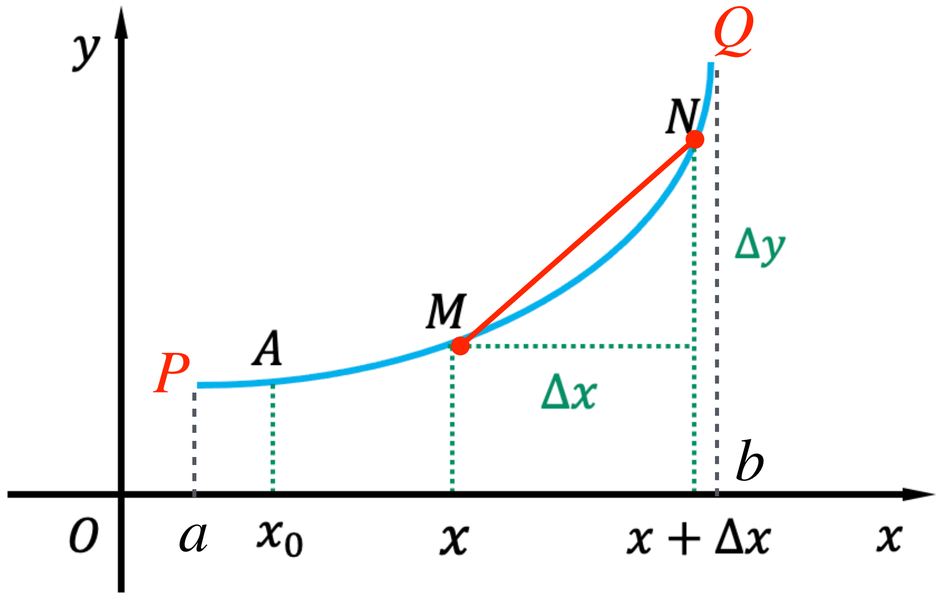

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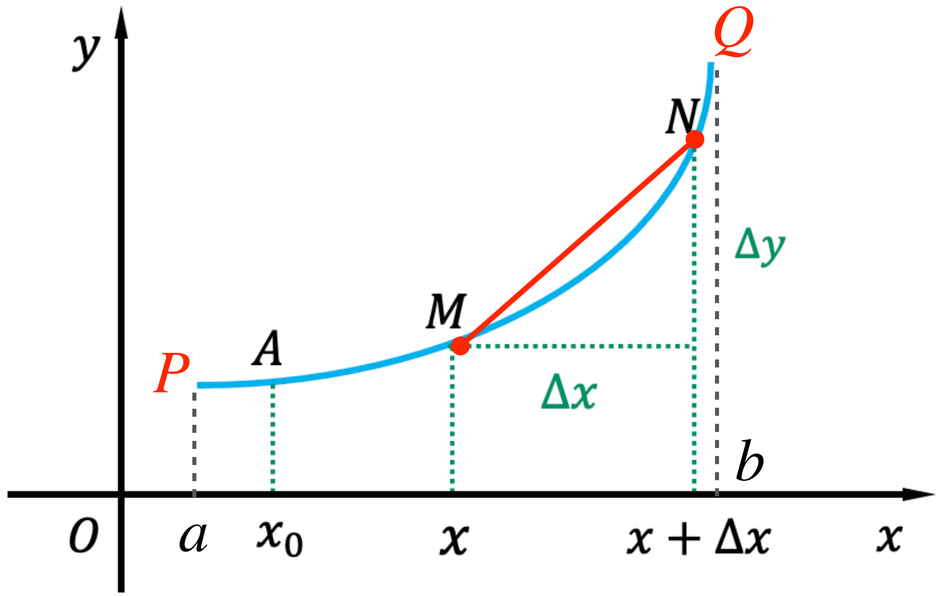

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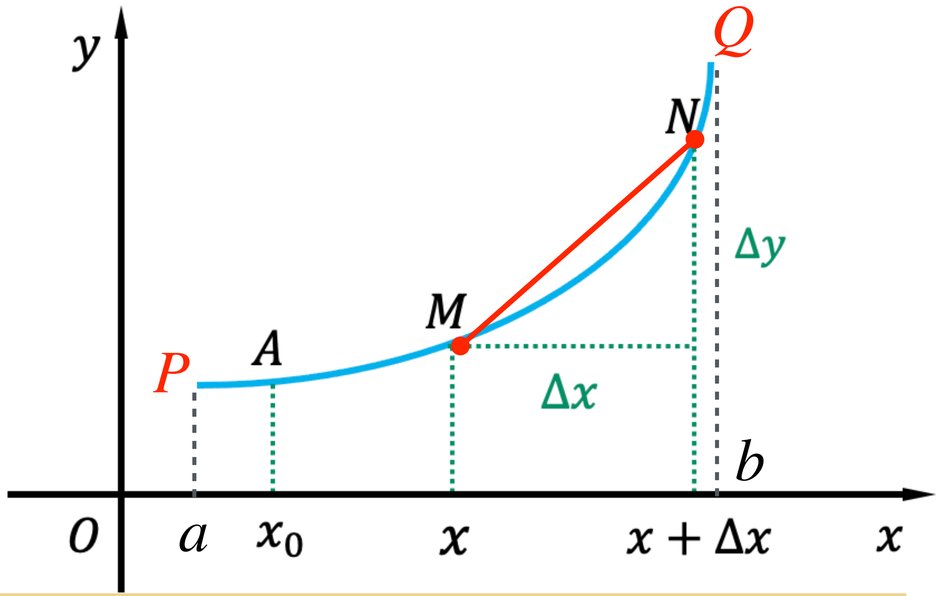

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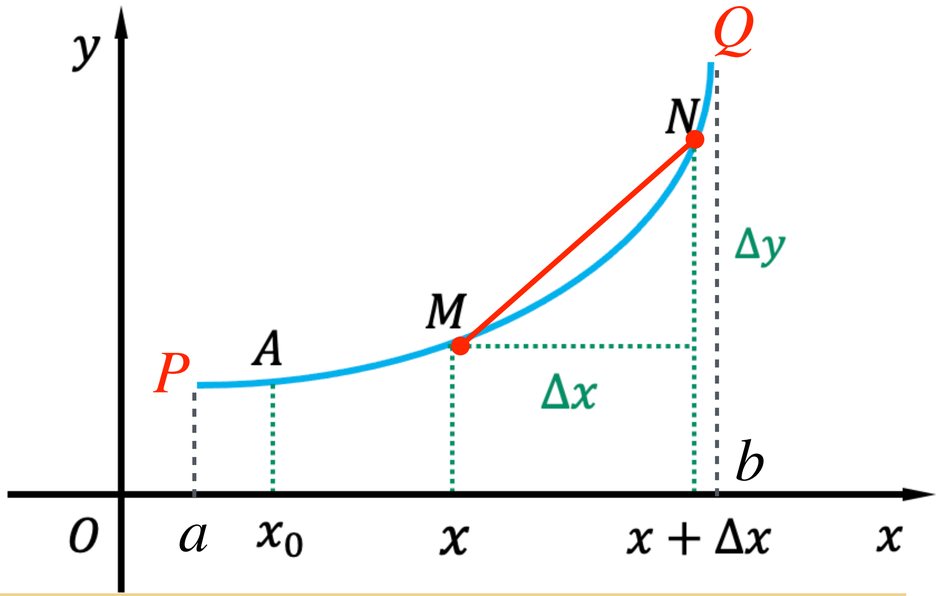

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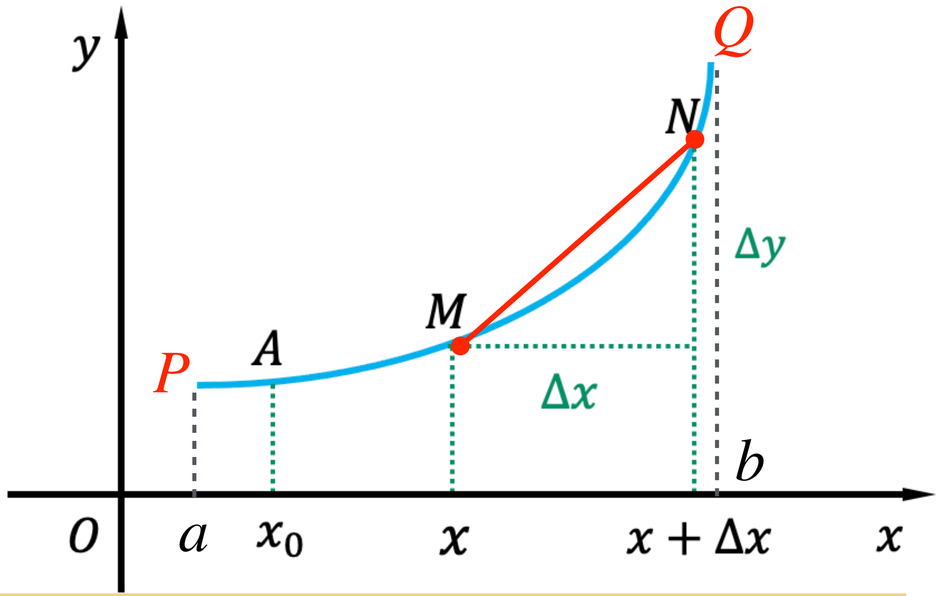

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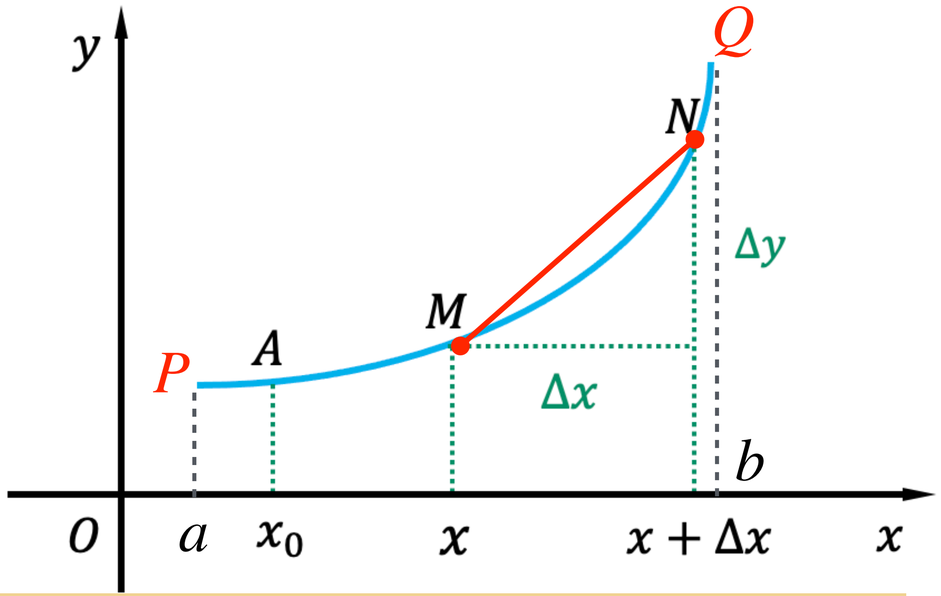

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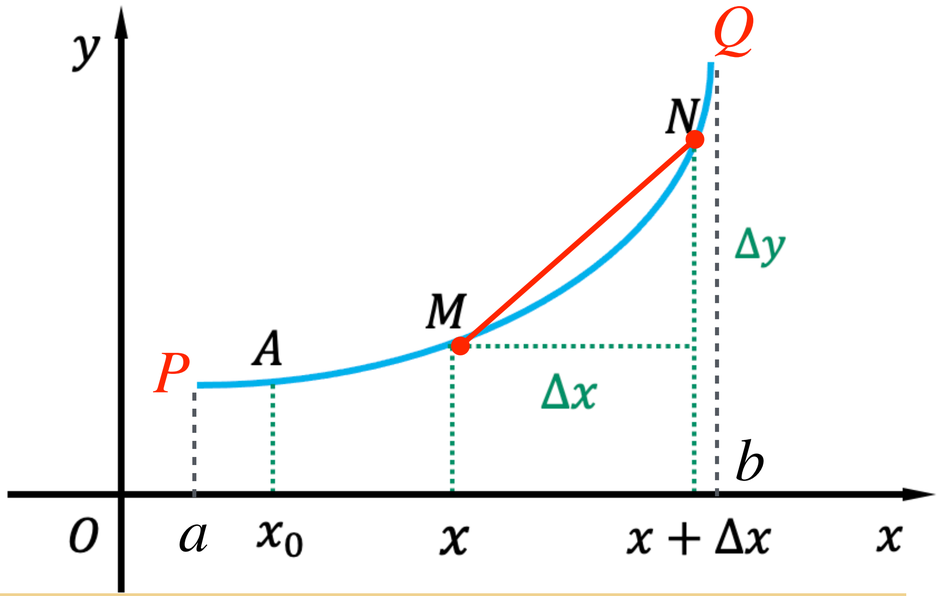

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弧微分公式

$$\mathrm{d}s = \sqrt{1 + \left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x$$

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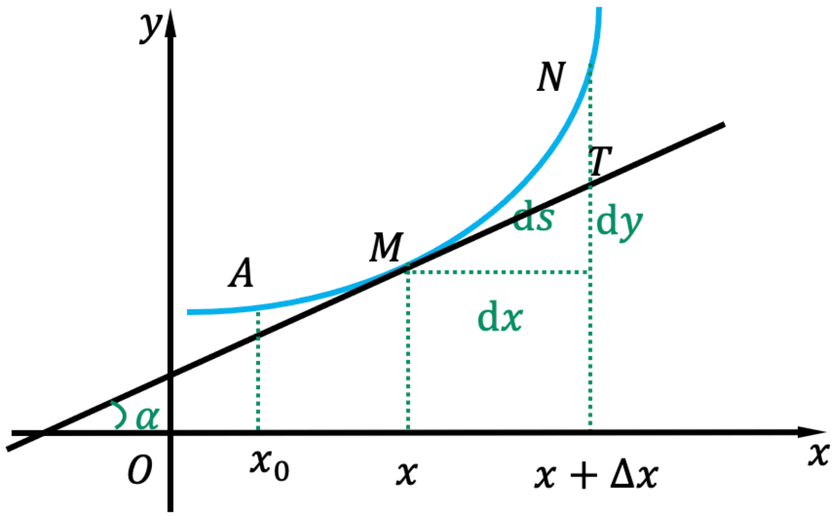
几何意义:

$$\mathrm{d}s = |MT|$$

$$\frac{\mathrm{d}x}{\mathrm{d}s} = \cos\alpha\,; \qquad \frac{\mathrm{d}y}{\mathrm{d}s} = \sin\alpha$$



弧微分

若曲线 $y=f(x)$ 换成参数函数 $\begin{cases} x=\varphi(t), \\ y=\psi(t) \end{cases}$ 其中 $\alpha \le t \le \beta$，且 $\varphi(t)$ 是单调函数，且

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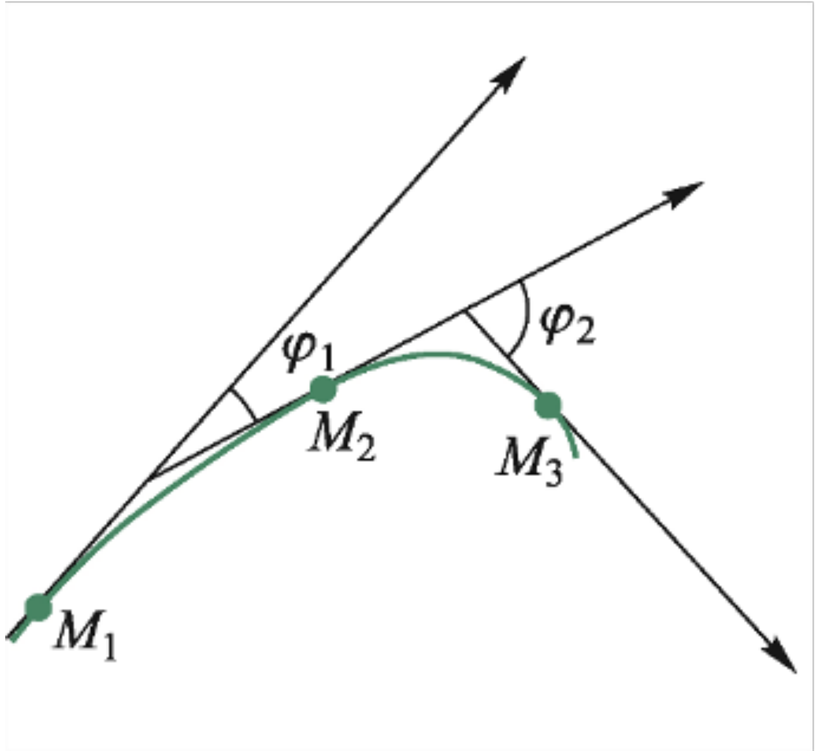
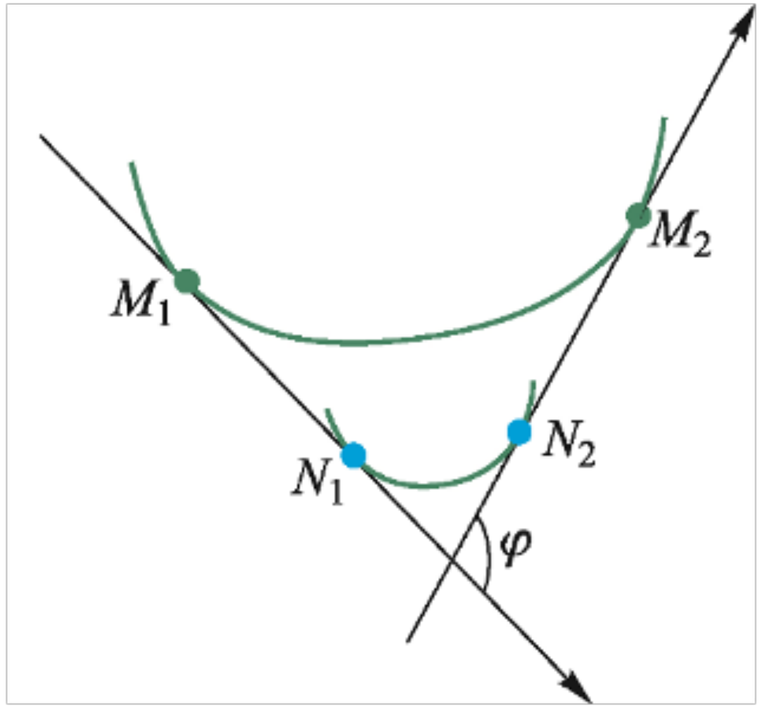
$$\mathrm{d}s=\sqrt{(\mathrm{d}x)^2+(\mathrm{d}y)^2}=\sqrt{\varphi'(t)^2+\psi'(t)^2}\mathrm{d}t$$

曲率

曲率是描述曲线局部性质（弯曲程度）的量.



弧段弯曲程度越大转角越大



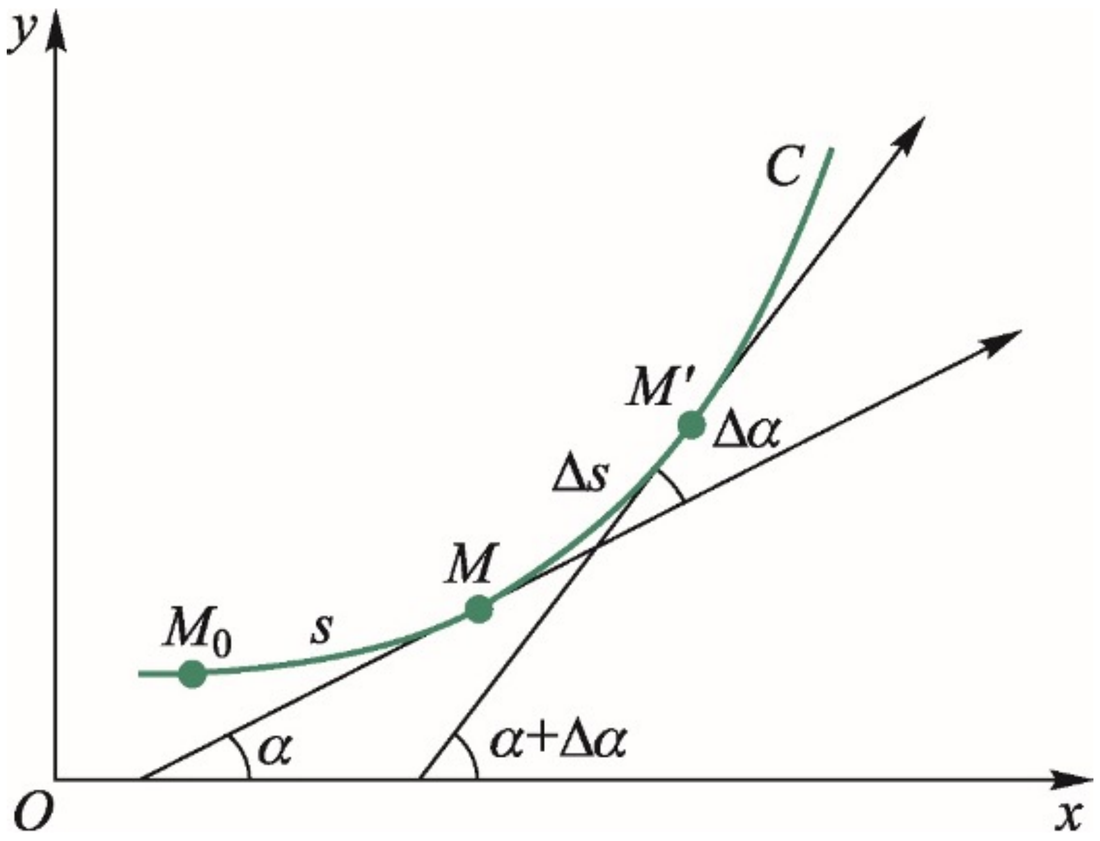
转角相同弧段越短弯曲程度越大

曲率

设曲线C是光滑的, M_0是基点.

$\left|\widehat{MM'}\right| = |\Delta s|$,

$M \to M'$ 切线转角为$|\Delta\alpha|$.



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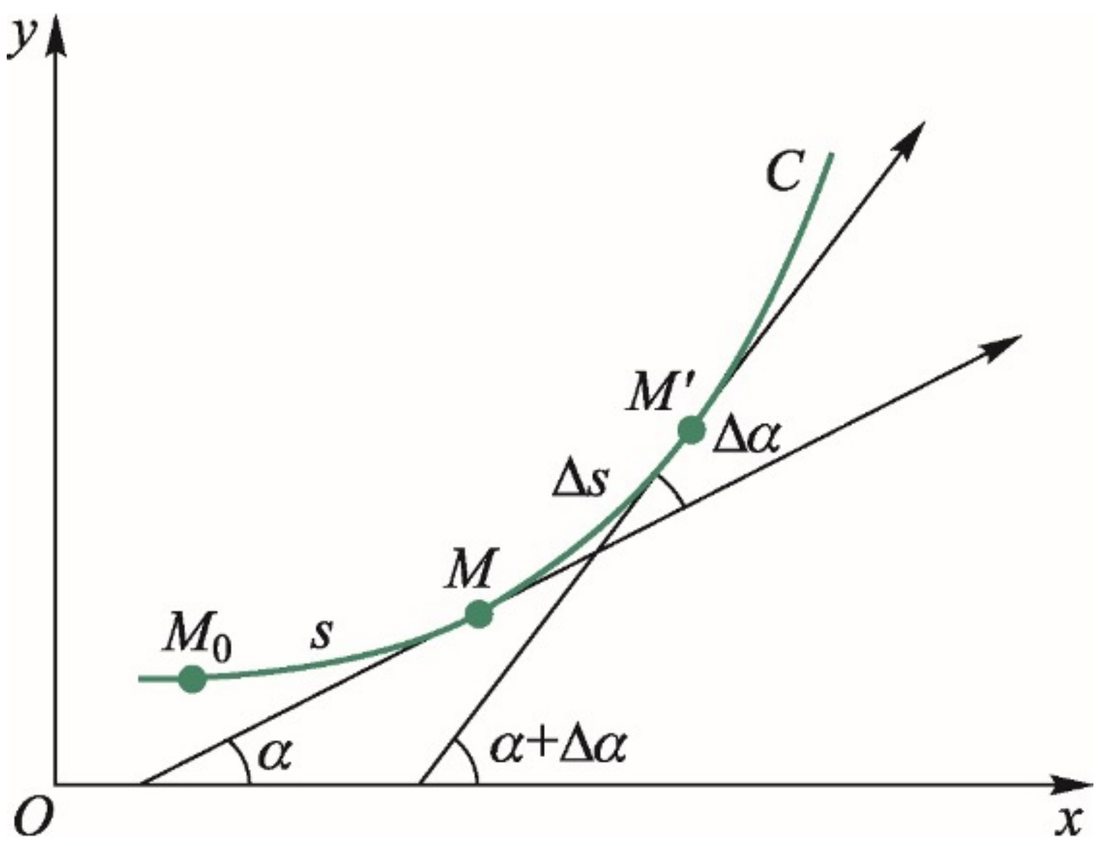
定义

弧段Δs上的平均曲率为$\bar{K} = \left|\dfrac{\Delta\alpha}{\Delta s}\right|$.

点M处的曲率为$K = \lim\limits_{\Delta s \to 0}\left|\dfrac{\Delta\alpha}{\Delta s}\right| = \left|\dfrac{\mathrm{d}\alpha}{\mathrm{d}s}\right|$.



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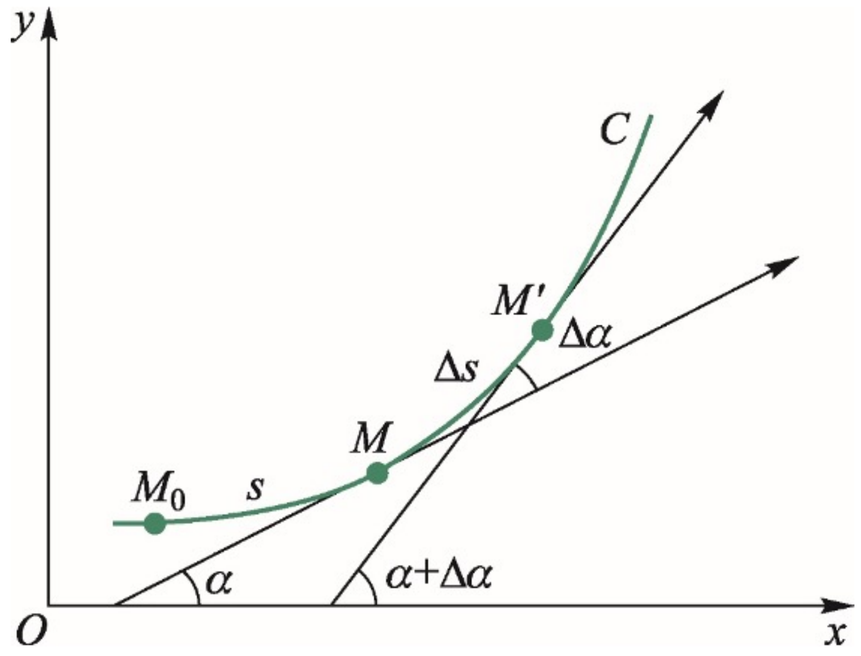


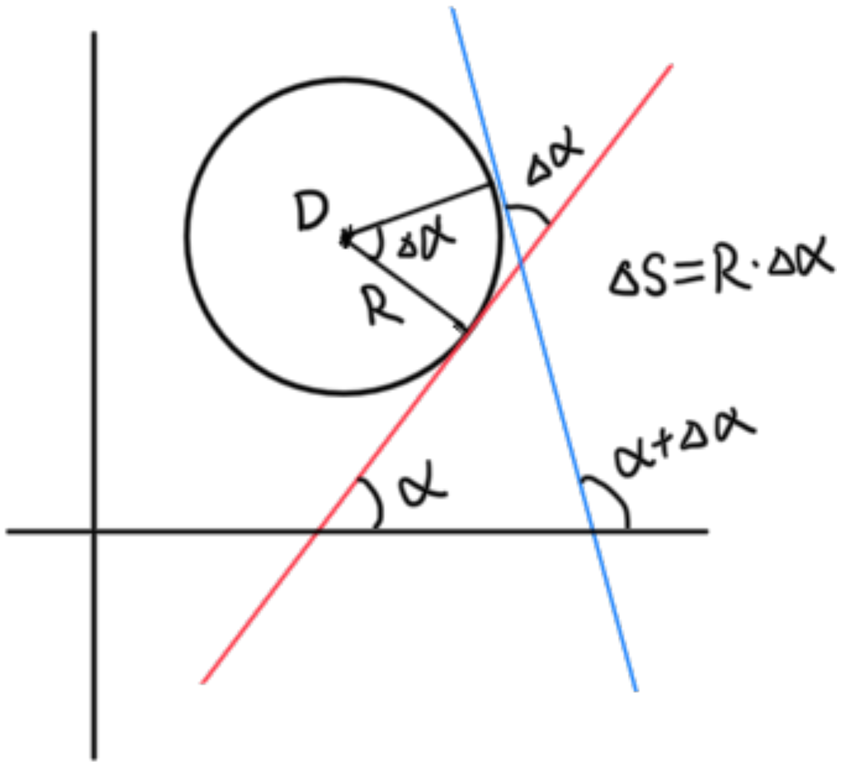
对于直线来说，任意点处的切线与它本身重合

所以当点沿直线运动时，切线的倾角的改变:

$\Delta\alpha = 0$, 从而 $K = \left|\dfrac{\mathrm{d}\alpha}{\mathrm{d}s}\right| = 0$

这与直线不弯曲相符合.

曲率

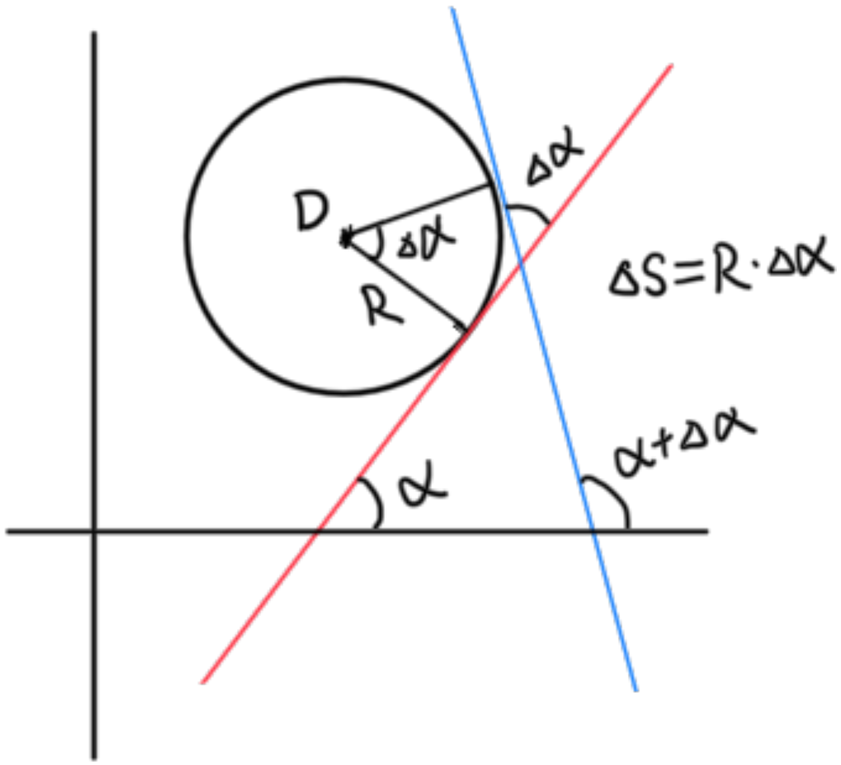


对于半径为 R 的圆，任意一点的曲率都相同：

$$K=\lim_{\Delta s\to 0}\left|\frac{\Delta\alpha}{\Delta s}\right|=\lim_{\Delta s\to 0}\left|\frac{\Delta\alpha}{R\cdot\Delta\alpha}\right|=\frac{1}{R}$$

、

曲率



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半径越小，曲率越大

半径越大，曲率越小

曲率

计算曲率的公式

设曲线的方程为 $y=f(x)$，且 $f(x)$ 具有二阶导数(此时 $f'(x)$ 连续，曲线光滑)

而切线的倾斜角 α，切线的斜率 k 与导数 y' 有下面的关系：

$$\tan\alpha = k = y' = \frac{\mathrm{d}y}{\mathrm{d}x}$$

故 α 是关于 x 的函数：

$$\alpha = \arctan y'(x)$$

所以：

$$\mathrm{d}\alpha = \frac{y''}{1+(y')^2}\mathrm{d}x$$

又

$$\mathrm{d}s = \sqrt{1+\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2}\,\mathrm{d}x = \sqrt{1+(y')^2}\,\mathrm{d}x$$

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$$K = \left|\frac{\mathrm{d}\alpha}{\mathrm{d}s}\right| = \frac{|y''|}{(1+(y')^2)^{3/2}}$$

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曲率计算公式

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曲率

如果曲线是由参数方程 $\begin{cases} x = \varphi(t), \\ y = \psi(t) \end{cases}$ 确定，其中 $\alpha \le t \le \beta$，且 $\varphi(t)$ 是单调函数，且

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则由参数方程求导得:

$$\frac{\mathrm{d}y}{\mathrm{d}x} = \frac{\psi'(t)}{\varphi'(t)} \qquad \frac{\mathrm{d}^2y}{\mathrm{d}x^2} = \frac{\varphi'(t)\psi''(t) - \varphi''(t)\psi'(t)}{\varphi'(x)^3}$$

故曲线的曲率的计算公式为:

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$$K = \left|\frac{\mathrm{d}\alpha}{\mathrm{d}s}\right| = \frac{|y''|}{(1+(y')^2)^{3/2}} = \frac{|\varphi'(t)\psi''(t) - \varphi''(t)\psi'(t)|}{\left[\varphi'(t)^2 + \psi'(t)^2\right]^{3/2}}$$

曲率

例1 计算等边双曲线$xy=1$在点(1,1)处的曲率.

例2 抛物线 $y=ax^2+bx+c$ 上哪一点的曲率最大?

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例1 计算等边双曲线$xy=1$在点(1,1)处的曲率.

解 $\because y'=-\dfrac{1}{x^2},\ y''=\dfrac{2}{x^3},$

$$\therefore\ K=\left.\frac{|y''|}{(1+y'^2)^{\frac{3}{2}}}\right|_{x=1,y=1}$$

$$=\frac{2}{[1+(-1)^2]^{\frac{3}{2}}}=\frac{\sqrt{2}}{2}.$$

例2 抛物线 $y=ax^2+bx+c$ 上哪一点的曲率最大?

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例2 抛物线 $y=ax^2+bx+c$ 上哪一点的曲率最大?

解 $\because y'=2ax+b,\quad y''=2a,$

$$\therefore\ K=\frac{|2a|}{[1+(2ax+b)^2]^{\frac{3}{2}}}.$$

显然, 当$x=-\dfrac{b}{2a}$时, K最大.

即抛物线在顶点处的曲率最大.

$$\boxed{\left(-\frac{b}{2a},-\frac{b^2-4ac}{4a}\right)}$$

曲率

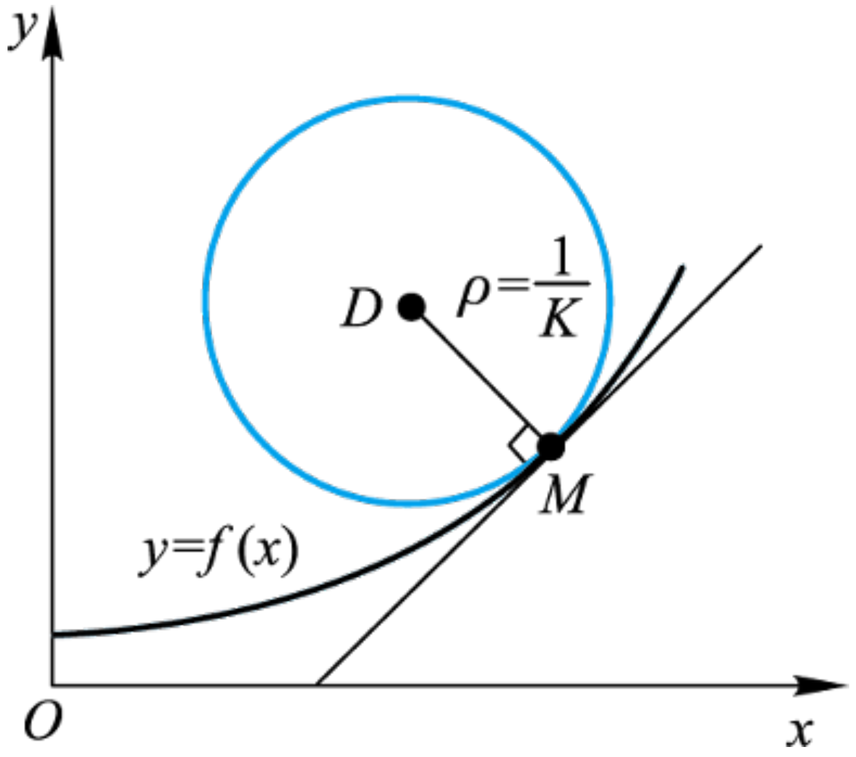
当 $|y'| << 1$ 时,有曲率近似计算公式$K = \dfrac{|y''|}{(1+y'^2)^{\frac{3}{2}}} \approx |y''|$

曲率圆与曲率半径

设曲线 $y=f(x)$ 在点 $M(x,y)$ 处的曲率为 $K(K\neq 0)$，在点 M 处的曲线的法线上，在凹的一侧取一点 D，使得

$$|DM|=\frac{1}{K}=\rho$$

以 D 为圆心，以 ρ 为半径作图（如左图），称此圆为曲线在点 M 处的曲率圆，D 称为曲率中心，ρ 称为曲率半径.



曲率圆与曲率半径

注意:

(1)曲线上一点处的曲率半径与曲线在该点处的曲率互为倒数.

$$\text{即 } \rho = \frac{1}{K}, \quad K = \frac{1}{\rho}.$$

(2)曲线上某点处的曲率圆与曲线有如下密切关系:

① 有公切线; ② 凹向一致; ③ 曲率相同 .

(3)曲线上一点处的曲率圆弧可近似代替该点附近曲线弧

(称为曲线在该点附近的二次近似).

曲率圆与曲率半径

例3 设工件内表面的截线为抛物线$y = 0.4x^2$. 现要用砂轮磨削其内表面, 问选择直径多大的砂轮比较合适?

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解 $\because y' = 0.8x, \quad y'' = 0.8.$

由例2可知, 抛物线在顶点(0,0)处曲率最大,

$$\therefore \ K = \left.\frac{|y''|}{(1+y'^2)^{\frac{3}{2}}}\right|_{x=0,y=0} = 0.8, \ \therefore \ \rho = \frac{1}{K} = 1.25.$$

抛物线顶点处的曲率半径

$\therefore$ 选用砂轮的半径不得超过1.25单位长, 即直径不得超过2.50单位长.